

A
PROJECT REPORT
ON
GPS Guided Autonomous Delivery Vehicle
Submitted For

EC253: Sensor Technologies

Submitted to
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Chapter 1

Abstract

This project details the design and implementation of the NITG In Campus Robo Delivery System, a fully autonomous, GPS guided wheeled robot engineered for last mile deliveries between fixed waypoints across a university campus. The system architecture replaces manual parcel and document transit by allowing users to dispatch the robot via a mobile web interface hosted directly on the microcontroller. Operating on an ESP32 WROOM-32 dual core processor, the robot navigates utilizing a NEO-6M GPS module for global positioning and an I2C magnetometer for precise heading correction. To ensure safe transit, a three channel ultrasonic sensor array provides real time obstacle detection. The firmware is strictly built upon a non blocking C++ state machine, utilizing a sequential waypoint queuing system to successfully navigate complex real world geometries such as V shaped roads, U turns, and curved walkways without human intervention.

Chapter 2

Introduction

Campus logistics, specifically the transit of documents, parcels, and inter department packages, traditionally rely on manual human trips between buildings. This project addresses the inefficiency of manual last mile delivery by automating the process through an autonomous robotic platform. By integrating modern spatial sensors and embedded web technologies, the DeliveryBot NITG provides a contactless, on demand delivery solution that operates entirely independently of remote control.

2.1 Objectives

1. To design a physical 2WD differential drive prototype capable of autonomous navigation between predefined campus waypoints (e.g., Terekhol Hostel, Admin Block, Gyan Mandir).
2. To implement a non blocking C++ state machine architecture that eliminates processing delays and ensures real time sensor responsiveness.
3. To develop a mobile first web User Interface (UI) hosted via an ESP32 Access Point for intuitive destination selection and dispatching.
4. To integrate a robust sensor suite combining GPS tracking and magnetic heading correction to overcome the limitations of straight line GPS navigation on curved campus pathways.
5. To deploy a sequential ultrasonic array that performs dynamic obstacle detection while actively preventing sensor crosstalk.

2.2 Components Used

- **Microcontroller (ESP32 WROOM-32):** The central dual core processing unit running the Arduino framework. It manages the HTTP server (AP Mode), state machine logic, and hardware abstraction.



Figure 2.1: Microcontroller (ESP32 WROOM-32):

Global Positioning System (NEO-6M): Interfaces via UART to provide 1Hz geographical coordinate fixes for macro navigation.

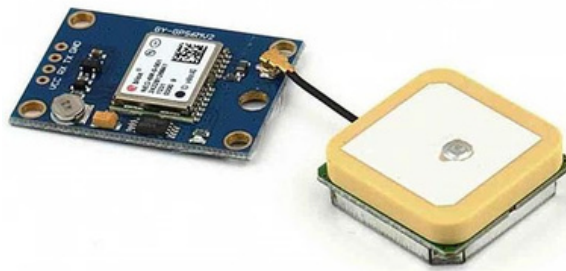


Figure 2.2: Global Positioning System (NEO-6M)

Magnetometer / Compass (HMC5883L/QMC5883L): Interfaces via I2C to provide continuous 8 point cardinal direction output for real time heading correction.



Figure 2.3: Magnetometer / Compass (HMC5883L/QMC5883L)

Ultrasonic Sensor Array (3x HC-SR04): Positioned at the Left, Center, and Right of the chassis for spatial awareness and obstacle detection.



Figure 2.4: Ultrasonic Sensor Array (3x HC-SR04)

Motor Driver (L298N): Governs the 2WD differential chassis via PWM signals. It utilizes an onboard 5V regulator to power the ESP32 logic, eliminating the need for a separate buck converter.

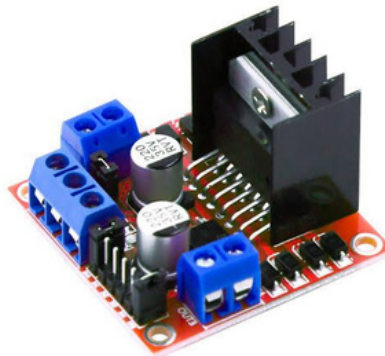


Figure 2.5: Motor Driver (L298N)

Drive Chassis: A 2WD differential wheeled platform allowing independent left and right motor speed control for precise maneuvering.



Figure 2.4: Drive Chassis

Chapter 3

Design and Implementation

3.1 System Overview:

The architecture of the DeliveryBot is segmented into three distinct software and hardware layers: the Mobile Web Interface, the Navigation and Control Core, and the Hardware Abstraction Layer. Power distribution is streamlined; the main battery routes directly into the L298N motor driver, which leverages its internal 5V regulator to safely supply the ESP32 logic board and connected sensors. The entire software stack operates under strict non blocking principles, meaning no `delay()` functions are permitted, ensuring the robot can simultaneously serve the web UI, read GPS bytes, and poll ultrasonic echoes without freezing.

3.2 Key Functional Block:

3.2.1 Central Processing Unit (ESP32)

- **Role:** Acts as both the web server and the navigation brain. The ESP32 creates a local WiFi network (SSID: DeliveryBot_NITG) hosted at 192.168.4.1. The UI is stored in PROGMEM with Base64 encoded assets, allowing users to select a destination securely and trigger the dispatch sequence.

3.2.2 Spatial Navigation Subsystem (GPS + Compass)

- **Role:** Standard GPS logic dictates moving in a straight line between two points, which fails on curved campus roads. This subsystem utilizes a Waypoint Queue. Pre surveyed coordinates form a breadcrumb trail. The NEO-6M verifies arrival at an intermediate node, while the magnetometer constantly corrects the chassis orientation to point toward the next specific coordinate in the queue.

3.2.3 Obstacle Detection Subsystem (Ultrasonic Array)

- **Role:** Three HC-SR04 sensors provide a forward facing spatial sweep. To prevent ultrasonic signals from bouncing and triggering false positives in adjacent sensors (crosstalk), the firmware enforces sequential firing. The state machine reads these values continuously; if an obstacle breaches the distance threshold, the robot transitions to a halt state.

3.2.4 Drive and Locomotion Subsystem

- **Role:** The L298N accepts variable Pulse Width Modulation (PWM) signals generated by the ESP32's LEDC peripheral. By manipulating the duty cycle independently for the left and right motors, the state machine can execute pivot turns, smooth arcs, and straight line acceleration phases.

Chapter 4

Operations

The operational pipeline of the digital transceiver represents a complete cycle of signal acquisition, digital conversion, RF transmission, and acoustic reconstruction. The system executes the following disThe operation of the autonomous delivery robot follows a strict sequential phase structure governed by its internal state machine:

Phase 1: Initialization and UI Serving Upon boot, the ESP32 initializes all hardware peripherals and performs a silent diagnostic check (verifying I2C lines and UART streams). It broadcasts its AP WiFi network. A user connects via a smartphone, accesses the captive portal, and selects a predefined surveyed location (such as the Seminar Hall or Main Gate) from the HTML interface.

Phase 2: Waypoint Queuing and Dispatch Once a destination is submitted, the `handleDispatch()` function is triggered. The system loads an array of intermediate GPS coordinates specific to that route. This queue ensures the robot follows the physical geometry of the campus roads rather than attempting to drive through buildings in a straight line.

Phase 3: Autonomous Transit The robot engages the drive motors. During transit, the non blocking loop continuously drains bytes from the NEO-6M GPS module to calculate distance to the current active waypoint. Simultaneously, it reads the magnetometer to calculate the required steering angle, adjusting the left and right motor PWM values dynamically to maintain the correct heading.

Phase 4: Real Time Obstacle Avoidance While moving, the ultrasonic array cycles continuously. If an obstacle is detected in the robot's path, the state machine instantly overrides the drive commands and halts the motors. In the current iteration, the robot waits for the obstacle to clear and retries the path. (Dynamic spatial re routing around the obstacle is marked as a deferred upgrade for future versions).

Phase 5: Arrival and Standby Once the GPS module verifies that the robot has reached the final coordinate in the queue within an acceptable margin of error, the system halts the motors, updates the web UI to reflect a successful delivery, and enters a standby state awaiting the next dispatch command.

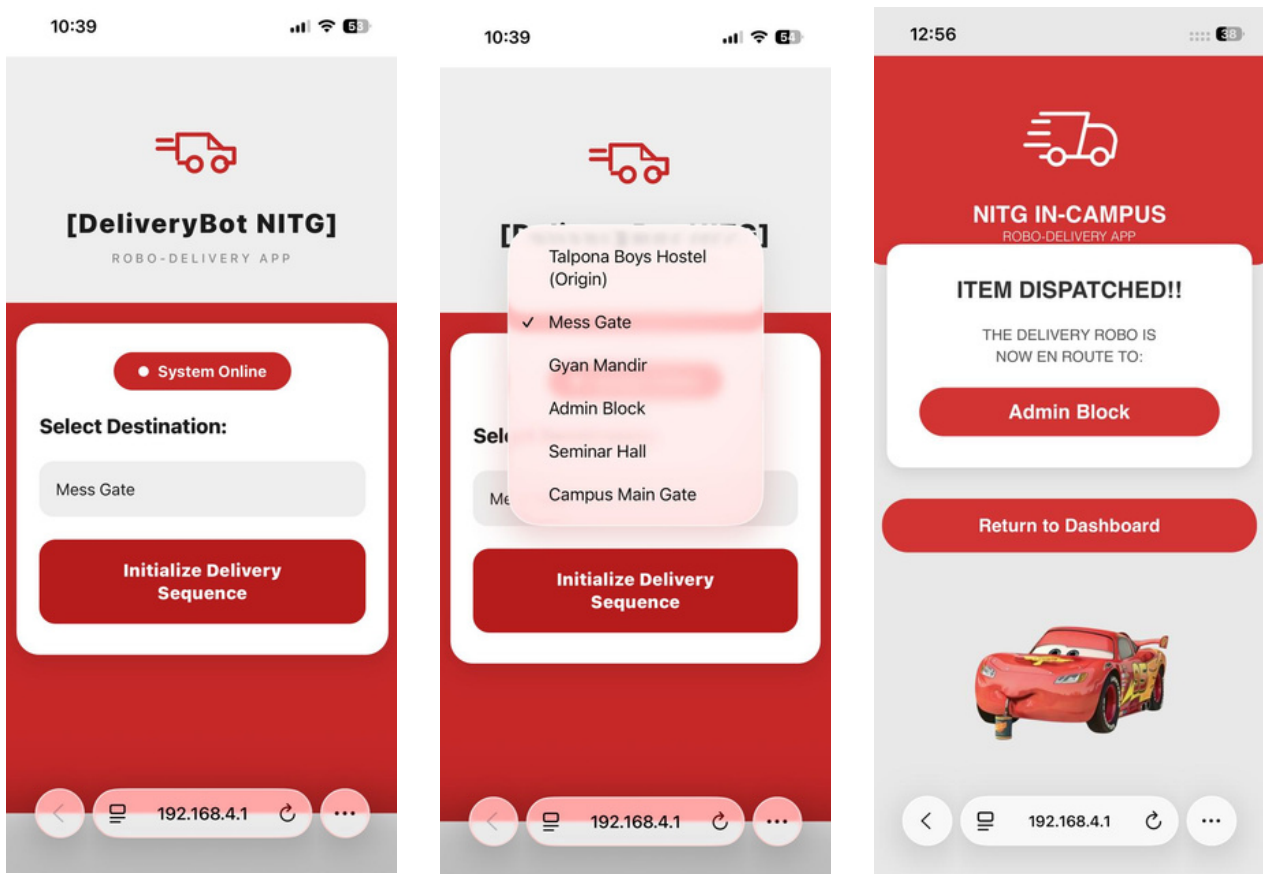
Chapter 5

Project Implementation

Code: ".ino" File Drive Link

Project Demonstration: Demonstration Video Link

User Interface:



Chapter 6

Cost Structure

Current Expense:

Subsystem	Specific Component	Estimated Cost (₹ INR)
Compute & Logic	ESP32-WROOM-32 Development Board	₹400
Navigation (Position)	U-blox NEO-6M GPS Module (with ceramic antenna)	₹400
Navigation (Heading)	QMC5883L (or HMC5883L) 3-Axis Magnetometer	₹150
Perception (Obstacles)	HC-SR04 Ultrasonic Sensor	₹80
Drive Control	L298N Dual H-Bridge Motor Driver	₹150
Actuation	2x 12V DC Gear Motors + Rubber Wheels	₹500
Power Supply	12V Battery Pack (Standard SLA or basic Li-ion)	₹600
Chassis & Hardware	2WD Chassis plate, caster wheel, standoffs	₹300
Misc. Electronics	Jumper wires, breadboard, 1k Ω & 2k Ω resistors (for the ECHO voltage divider)	₹150
Total Current Cost	DeliveryBot Prototype	~₹2,730

Chapter 7

Results and Conclusion

5.1 Results

1. **Firmware Stability:** Version 4.0 of the firmware was successfully deployed, resolving critical logic bugs including the elimination of hidden 90ms blocking delays in the ultrasonic read loop and fixing LEDC API incompatibilities between ESP32 Core versions.
2. **Hardware Validation:** Individual validation sketches successfully proved the isolated functionality of the 5 phase motor PWM ramping, sequential ultrasonic firing, raw I2C compass auto detection, and non blocking GPS byte drainage.
3. **Navigational Accuracy:** The implementation of the waypoint queue system successfully corrected previous navigational failures, allowing the chassis to handle V shaped roads, complex curves, and real world GPS drift without requiring manual steering intervention.

5.2 Applications

1. **Campus Logistics:** Automated, contactless transport of library books, administrative documents, or lab materials between university departments.
2. **Medical Facilities:** Safe transportation of non critical medical supplies or sterile equipment across hospital wings without tying up human staff.
3. **Corporate Tech Parks:** Last mile parcel delivery within closed, geofenced corporate environments.

5.3 Conclusion

This project successfully demonstrates the integration of multiple distinct sensor technologies into a cohesive, autonomous robotic platform. By strictly adhering to non blocking software architecture and leveraging an ESP32 as both an embedded web server and a localized navigation brain, the system bridges the gap between hardware abstraction and practical logistics. The DeliveryBot prototype satisfies the core objectives of spatial awareness, waypoint navigation, and user accessible dispatching, proving the viability of autonomous robotic transit within complex, structured environments.